Knowledge Graphs and the Semantic-Model Perspective

Two NMOP knowledge-graph drafts read alongside "Meaning Across Representations"

A neutral comparative reading — prepared for discussion

PURPOSE

This note reads two recent NMOP Internet-Drafts — a knowledge-graph framework for network operations (draft-mackey-nmop-kg-for-netops) and an incident-management knowledge graph (draft-tailhardat-nmop-incident-management-noria, "NORIA") — alongside the discussion deck "Meaning Across Representations." The aim is neither to endorse nor to rank the three, but to lay them side by side and draw out where they align and where they genuinely differ, so that a reader navigating this area can see how the conceptual argument in the deck and the concrete architectures in the drafts relate. The deck already lists the knowledge-graph work in its own survey of the field, under semantic annotation and knowledge graphs, so the three are not strangers to one another; the value here is in making the correspondences and the divergences explicit.

1. The three documents in brief

"Meaning Across Representations" (the deck). A conceptual synthesis. It argues that a number of separate IETF and IRTF conversations — AI-consumable YANG, ontology reconciliation, reference lexicons, NAIM, multi-vendor normalisation, semantic annotation and knowledge graphs, and the ONSEN YANG↔TMF bridge — are takes on a single need: a semantic model, meaning a model's meaning (its ontology, lexicon and use, together with provenance) made explicit and independent of form. Its central claims are that reconciliation across two representations is inherently cross-layer, that a practical semantic model is therefore built ad hoc for each bridge rather than pre-agreed as a universal artefact, and that the useful research programme is to automate that ad hoc creation, standardize only thin supports, and publish reusable pieces.

draft-mackey-nmop-kg-for-netops. An architectural proposal. It positions an RDF-based knowledge graph as a layer that connects data across the management, control and data planes — YANG, IPFIX, BMP, syslog, TMF and more — using RDFS/OWL, SPARQL, SHACL and resolvable IRIs. It is explicit that this is "not a replacement for YANG, but a complementary layer," and that its authors "are not proposing creating a new model," but rather ways to connect existing data and, through those connections, recover its meaning. It sketches an agent-based closed loop and reports an early prototype (a yang2rdf tool).

draft-tailhardat-nmop-incident-management-noria (NORIA). An architectural proposal focused on incident management. It builds an IT-Service-Management knowledge graph from a layered ontology: YANG models lifted automatically into RDFS/OWL (ONTO-YANG-MODEL), an abstract service-management ontology (ONTO-META, exemplified by NORIA-O), and a linking layer (ONTO-LINKER) that records equivalences between them.

Heterogeneous models are reconciled by alignment — for instance owl:equivalentClass relations queried through federated SPARQL — “without requiring a priori universal consensus.” It reaches deliberately beyond YANG’s scope into physical, organizational and business context, and emphasises shareable “incident signatures.”

2. At a glance

The table condenses the comparison; the sections that follow discuss the alignments and differences in turn.

Dimension	Meaning Across Representations (deck)	draft-mackey KG-for-NetOps	draft-tailhardat NORIA
Genre & intent	Conceptual synthesis and research framing; many threads, one need.	Architecture + call to standardize connective formats; early prototype.	Architecture for incident management; tooling and the NORIA-O ontology.
Claim about models	Meaning made explicit and form-independent; a semantic model bridges. Not a universal model.	KG is “a complementary layer,” “not a replacement for YANG”; no new model.	KG layer over YANG; transform to RDF/OWL and embed in a broader model.
What “meaning” is	Semantics + pragmatics: ontology, lexicon and use — with provenance; excludes form.	Semantics recovered from connections among data; ontologies as scaffolding.	Ontology (RDFS/OWL): concepts and relationships; extends to ITSM, org, location.
Technology	Stack-neutral; conceptual (the semiotic triad).	Semantic web: RDF, RDFS/OWL, SPARQL, SHACL, IRIs.	Semantic web: RDFS/OWL, federated SPARQL; layered ontologies.
Reconciliation	Cross-layer, built ad hoc per bridge; no single layer reconciles alone.	Per-case: materialize or virtualize; declarative mappings; overlay triples.	Alignment via owl:equivalentClass; federated SPARQL; no a-priori consensus.
What to standardize	Thin supports only: reference lexicons and provenance conventions.	Import formats (→RDF), deterministic IRIs, declarative mapping languages.	YANG→OWL transformation; linking ontologies; shared ITSM access.
Reuse / commons	Publish correspondences per case; bind N to a reference, not N^2 pairs.	Shared knowledge base; resolvable IRIs enable federation and reference.	Shareable “incident signatures” without exposing sensitive detail.
Active process & dynamics	Cognitive agents reconcile at runtime; companion work adds negotiation of in-life change.	Agent-based closed loop; inference engine “split into agents” over the KB.	Mostly offline analysis and knowledge capitalization; limited closed loop.
Scope of world	Network information representation and boundary-crossing.	Cross-plane operational silos: intent ↔ state ↔ inventory.	Incident management incl. physical, organizational, ITSM context.
Maturity / venue	Discussion deck; NMRG-oriented; two janz drafts.	NMOP I-D; concepts plus prototype.	NMOP I-D; tooling plus NORIA-O ontology.

3. Where they align

The strongest alignment is the foundational one: all three decline the universal model. The deck argues that independently-built, evolving systems “differ in scope and ontology, so meaning can't be settled in full in advance," and that only thin supports should be standardized. Mackey's draft states almost the same thing operationally — the knowledge graph is a complementary layer, not a replacement, and the authors are "not proposing creating a new model." NORIA reconciles by alignment "without requiring a priori universal consensus." Three documents, written from different starting points, converge on the same architectural refusal.

They agree, too, that meaning lives above form and must be made explicit. The deck's whole premise is that machines have long consumed form and the new demand is the layers above it — semantics and pragmatics. The knowledge-graph drafts are, in effect, a way of doing exactly that: lifting YANG and other schemas into a representation in which relationships and meaning, not just structure, can be expressed and queried. Where the deck says "make models' meanings explicit," the drafts say "import the schemas into RDF and overlay the relationships," which is one concrete method for the same move.

Reconciliation, in all three, is per-case rather than pre-agreed. The deck insists it is cross-layer and built for the bridge at hand; Mackey materializes or virtualizes sources and overlays mapping triples case by case; NORIA aligns YANG-derived ontologies to an abstract one through explicit equivalence links. None assumes that two systems' concepts line up in advance; each provides machinery for settling the correspondence when it is needed.

They also share the instinct to standardize only thin things, and to let the results be reused. The deck names reference lexicons and provenance conventions as the standardizable supports and calls for a bottom-up commons of published correspondences. Mackey would standardize import formats, deterministic identifiers and mapping languages — supports to the connective process, not a master schema — and treats resolvable IRIs and a shared knowledge base as the reuse mechanism. NORIA's shareable incident signatures are the same reuse instinct in the incident-management domain. And all three give active computation a real role: the deck's cognitive agents that reconcile at runtime, Mackey's inference engine "split into individual agents" collaborating over the knowledge base, and NORIA's automated lifting and alignment tooling.

4. Where they differ

Genre and altitude. The clearest difference is not disagreement but level. The deck is a conceptual account: it argues why reconciliation is cross-layer, why a universal model is neither achievable nor desirable, and how the pieces of a research programme fit. The two drafts are architectures with tooling: they commit to specific components, data flows and formats, and report prototypes. The deck explains a shape; the drafts build one instance of it. Read together, the deck supplies reasons that the drafts largely assume, and the drafts supply concreteness the deck deliberately leaves open.

Technology commitment. The deck is stack-neutral by design — it speaks of a semantic model in the abstract, grounded in the semiotic triad, and does not require any particular representation. Both drafts commit to the semantic-web stack: RDF, RDFS/OWL, SPARQL, SHACL, IRIs. This is a genuine divergence of kind, not degree. It means the drafts inherit both the strengths of that stack (federation, standard query and validation, resolvable identity) and its commitments, whereas the deck's argument would apply equally to a different realization. A reader should not mistake the drafts' RDF machinery for a premise of the deck's case.

What counts as "meaning." The deck is explicit that meaning, in its broad sense, is semantics and pragmatics together — ontology, lexicon and use — and it makes provenance a named layer of the semantic model. The knowledge-graph drafts operationalize concepts and relationships thoroughly, and Mackey recovers "the semantic of the information" from connections; but the pragmatic dimension — use and context — and provenance are handled more implicitly, through identifiers, context metadata and, in NORIA, identifier-drift mitigation, rather than as first-class layers. The deck's account is wider in what it counts as meaning; the drafts are deeper in one part of it.

Where the bridge lives. The deck holds that the bridging construct is built per case and is not itself something to standardize. The drafts externalize correspondences as explicit artefacts — mapping triples, owl:equivalentClass alignments — and Mackey proposes to standardize the formats for expressing them. On a careless reading these look opposed, but they need not be: there is a difference between recording a correspondence, which is thin, reusable data an active element produces, and freezing the mapping function itself into a shared model. The drafts do the former and leave the determining to active components; the deck endorses exactly that, and calls the recorded results the reusable commons. The apparent tension resolves into a shared position, provided one keeps "publish the correspondence" distinct from "model the bridge."

Scope of the world modelled. The deck confines itself to the representation of network information and the crossing of model and system boundaries. NORIA reaches well beyond that, into physical location, organizational structure, business process and IT-service-management — because incident management needs that context. Mackey sits in between, spanning operational data across planes and the intent–state–inventory divide. The drafts therefore take on kinds of heterogeneity the deck does not explicitly treat, which is natural given their operational targets.

Dynamics and the treatment of change. Here the three separate most. The deck foregrounds runtime reconciliation, and its companion line of work adds negotiation of in-life change as a first-class process. Mackey's architecture is explicitly closed-loop, with agents acting over the knowledge base. NORIA, by contrast, concentrates on offline analysis and knowledge capitalization — anomaly detection, root-cause analysis, sharable signatures — with comparatively little closed-loop control. If one is reading specifically for how motion and in-life change are handled, Mackey and the deck's companion work engage it directly, while NORIA is more a static-knowledge instrument.

5. Reading the three together

Placed side by side, the drafts can be read as concrete existence proofs for the deck's central claim, arrived at independently: build the semantic layer ad hoc and federated, keep the base models thin, standardize only the connective supports, and let active components do the reconciling. That the two NMOP drafts reach this posture without reference to the deck’s argument is itself informative — it suggests the position is one practitioners converge on when they confront heterogeneous operational data, not a stance peculiar to one line of thought.

The genuine differences are worth keeping in view rather than smoothing over. The drafts commit to a technology stack the deck holds at arm’s length; they operationalize concepts and relationships more fully than they do use and provenance, which the deck treats as equal parts of meaning; and they divide on dynamics, with one architecture closed-loop and the other largely offline. None of these is a contradiction of the deck; each is a choice the deck leaves open being made one way. The instructive point is that the conceptual account and the concrete architectures occupy different altitudes of the same problem, and are most useful read as complements: the deck for why the shape is what it is, the drafts for what one built instance looks like and what it still leaves unsettled — notably the governance of shared ontologies and mappings, which both drafts raise as an open question and which the deck would answer as a matter of thin, shared support rather than universal agreement.

DOCUMENTS COMPARED

- Meaning Across Representations — discussion deck (this repository).
- draft-mackey-nmop-kg-for-netops — A Knowledge Graph Framework for Network Operations (IETF NMOP).
- draft-tailhardat-nmop-incident-management-noria — Incident Management for Network Operations / NORIA (IETF NMOP).